

Module 2 — Personalised Drug Exposure and Renal Study Planning

What the module does, how it is built, what data it uses, how well it works, and where it fails. Written for readers with an engineering background and no healthcare training. Every term is defined the first time it is used.

ML MODELS 0 equations, not learning	DRUGS COVERED 39	VALIDATION RECORDS 25 across 17 drugs	WITHIN 2× 88% standard acceptance bar	STUDY DESIGN CORRECT 16 / 17
COSTLY ERRORS 0 missed full studies				

01 Problem Statement

1.1 · THE WORDS YOU NEED FIRST

Kidney clearance

Some drugs leave the body through the kidneys, unchanged. The kidneys filter them out into urine.

Liver clearance

Other drugs are broken down by the liver into different chemicals before leaving the body.

Exposure (AUC)

The total amount of drug in the blood, added up over time. If the body removes a drug more slowly, the drug stays longer and the exposure is higher.

eGFR

A number describing how well the kidneys are working, calculated from a routine blood test. Around 90 or above is normal. Below 15 means the kidneys have essentially stopped.

fe

The fraction of a drug that leaves through the kidneys unchanged. A value of 0.90 means 90% leaves through the kidneys and 10% by other routes.

1.2 · WHAT GOES WRONG

Module 1 decides whether two drugs interact, and how serious that is. It answers that question for an average healthy adult.

Real patients are not average. Kidney function varies widely between people of the same age. When a patient's kidneys work at half their normal rate, a drug that leaves through the

kidneys is removed at half the rate — so it builds up in the blood. The dose printed on the box was chosen for someone whose kidneys work normally.

The result is that the same two drugs, at the same doses, can be safe in one patient and dangerous in another. If a clinical trial puts both kinds of patient into one group and reports the average, the danger to the second kind is hidden inside that average.

1.3 · THE TWO QUESTIONS MODULE 2 ANSWERS

QUESTION	WHO ASKS IT	WHAT MODULE 2 RETURNS
For this patient — how much does each drug build up, and does that change the risk of the interaction?	A clinical pharmacology team screening who can safely join a trial	A predicted exposure figure per drug, a revised risk level, and a written reason for every change
For this drug — what kind of kidney study does it need before trials begin?	The team planning the trial programme, and the committee funding it	Either "Full study" or "Reduced study", with the reason stated

1.4 · WHY THE SECOND QUESTION INVOLVES MONEY

Before a drug is approved, regulators expect the company to measure how kidney impairment changes that drug's exposure. There are two accepted designs:

- **Full study** — four separate groups of patients: normal kidney function, mild impairment, moderate impairment, and severe impairment.
- **Reduced study** — two groups only: normal function and severe impairment. If the severe group shows no meaningful change, nothing further is needed.

A reduced study recruits roughly half as many patients. Across our 39 drugs, Module 2 recommends a full study for 15 and a reduced study for 24.

THE ERROR THAT ACTUALLY MATTERS

The two mistakes are not equally expensive. Recommending an unnecessary full study wastes part of one study. Recommending a reduced study when the kidney genuinely matters means the problem is not found until much later in development — after far more money has been committed. Module 2 is measured on those two error types separately, not on a single accuracy score.

02 Tech Stack

2.1 · WHAT THE PRODUCT RUNS ON

COMPONENT	VERSION	WHY IT IS USED
Python	3.13	The language the whole module is written in
math (built in)	–	The only library the core calculation needs. The equations are arithmetic, powers and one logarithm
Streamlit	≥1.30	Builds the web page where the user types in the patient details and reads the result
Plotly	≥5.18	Draws the exposure chart
pandas	≥2.0	Arranges results into tables for display
NumPy	≥1.24	Used indirectly by pandas and Plotly
RDKit	≥2023.9.1	Handles chemical structures. Used by Modules 1 and 2 for the drug list, not by the exposure calculation

No paid services. No internet connection needed while running. No graphics card. The whole module runs on an ordinary laptop processor.

2.2 · EXTRA SOFTWARE USED ONLY FOR CHECKING, NOT FOR RUNNING

A separate set of tools is used to check Module 2's equation against detailed physics-based simulation. None of it is needed to use the product.

COMPONENT	VERSION	WHY IT IS USED
PK-Sim	12.3.173	Free, published simulation software that models the whole body organ by organ. Used as an independent reference
R	4.6.1	Runs the exported simulations without opening the PK-Sim window
ospsuite (R package)	12.4.5	Loads a saved simulation, runs it, and reports the exposure figure
rSharp (R package)	1.2.3	Connects R to the simulation engine. The version is fixed deliberately — newer versions break file loading

03 Models Used

STATE THIS PLAINLY IF ASKED

Module 2 contains no machine learning. Nothing is trained, nothing learns from examples, and there are no weights to fit. It is a set of published equations, a table of drug properties, and threshold rules. Calling it "AI" would be inaccurate.

3.1 · THE FOUR PUBLISHED EQUATIONS

These are standard equations from clinical pharmacology. We implemented them; we did not invent them.

EQUATION	WHAT IT CONVERTS	INTO
CKD-EPI 2021	A creatinine blood test result, plus age and sex	eGFR — a kidney function score. This is the current standard version, which does not use race as an input
Du Bois formula	Height and weight	Body surface area, in square metres
Absolute eGFR	eGFR and body surface area	Kidney function adjusted to the patient's actual body size. The standard eGFR number is scaled to a fixed reference body; drug removal depends on the real one
Cockcroft-Gault	Age, weight, sex, creatinine	An older kidney measure. Displayed for reference only, because some older drug labels state their limits in this unit. The calculation does not use it

3.2 · THE ONE EQUATION THAT IS OURS

```
KF = kidney function as a fraction of normal    = min(1, absolute eGFR / 90)
HF = liver function as a fraction of normal
fe = fraction of the drug removed by the kidney
fh = fraction removed by everything else        = 1 - fe

CL_ratio = fe x KF + fh x HF
AUC_ratio = 1 / CL_ratio
```

Read in words: the drug's total removal rate is the kidney part scaled by how well the kidneys work, plus the non-kidney part scaled by how well the liver works. If the body removes the drug at half the normal rate, the drug builds up to twice the normal level.

This is why the answer depends on which drug is being asked about. A drug with `fe = 0.90` in a patient with badly reduced kidney function builds up severely. A drug with `fe = 0.00` in

the same patient is barely affected, because the kidneys were never removing it.

3.3 · THE RULE TABLES

Beyond the equations, the module applies fixed rules:

RULE SET	CONTENTS	SOURCE
Kidney function categories	Normal (90+), Mild (60–89), Moderate (30–59), Severe (15–29), Kidney failure (below 15)	FDA 2024 renal impairment guidance
Exposure bands	Below 1.25× is treated as no meaningful change. 2× or above is a clinically important change	Standard regulatory practice
Study design rule	Full study if fe is 0.30 or above, or if severe impairment is predicted to reach 2× exposure. Otherwise reduced study	Our rule, following the structure of the FDA guidance
Named hazards	Nine specific situations, listed in section 3.4	Standard clinical pharmacology

3.4 · THE NAMED HAZARDS

These are situations where one property of a drug meets one weakness of a patient. Each one adds to the risk score and writes a sentence explaining why.

- A drug that can injure the kidney, given to a patient whose kidneys are already impaired.
- A drug that can injure the liver, given to a patient with reduced liver function.
- Metformin specifically, in a patient with eGFR below 30 — a stated contraindication.
- A sedating drug given to a patient aged 75 or over.
- A drug that raises blood potassium, in a patient whose kidneys cannot remove potassium.
- A drug that raises potassium, in a patient whose potassium is already high.
- A drug that affects heart rhythm, in a patient whose potassium is low.
- A drug with a narrow safety margin that also builds up in this patient.
- Reduced clearance generally, which scales the whole interaction risk upward.

The module also records reasons the risk is *lower* than average — for example, a patient under 60 with normal kidney and liver function.

04 Dataset

Module 2 uses three separate collections of data, for three different purposes.

4.1 · DRUG PROPERTIES — THE INPUT TO THE CALCULATION

A table of **39 drugs** in `trialsense/drugs.py`. For each drug it holds the chemical structure, the fraction removed by the kidney (`fe`), and a set of yes/no flags — whether it can injure the kidney, whether it affects heart rhythm, whether it has a narrow safety margin, and so on.

These values come from standard clinical pharmacology references. They are described in the code as approximate and rounded. The chemical structures are checked automatically: the code recalculates each molecule's weight and fails if it does not match the published value, so a typing error cannot reach the demo unnoticed.

4.2 · REAL PATIENT DATA — USED TO CHECK THE PREDICTIONS

The file `data/renal_validation.json` holds what actually happened to drug exposure in patients with impaired kidneys. It was collected from FDA drug labels and published studies.

Drugs with a record in the file	39
Drugs with a usable number	17
Observations scored	25
Records excluded, each with a stated reason	19
Labels stating plainly that kidney disease changes nothing	9

Every scored observation carries a word-for-word quote from its source and a link, so any number can be traced back. Records are excluded for stated reasons — for example, patients on dialysis are excluded because dialysis removes drug by a route the equation does not represent.

AN IMPORTANT DISTINCTION

This data is the **reference**, not a training set. It is never used to adjust the equation. If it were, reporting accuracy against it afterwards would be meaningless.

4.3 · SIMULATION MODELS — AN INDEPENDENT SECOND CHECK

Seven published whole-body simulation models were downloaded from the Open Systems Pharmacology project on GitHub: Fluconazole, Metformin, Methotrexate, Omeprazole,

Atorvastatin, Ketoconazole and Theophylline. Each is pinned to the exact version it was downloaded from, recorded with its web address.

These provide a second opinion from a completely different method. The real patient data in 4.2 is the primary reference; the simulations test whether the equation behaves correctly across kidney function levels where real data is thin.

4.4 • PATIENT DATA — NONE IS STORED

The patient details used by the module — age, weight, height, sex, and four routine blood test results — are typed in by the user when they use it. Nothing is saved, and no patient records are held anywhere in the project.

All seven inputs are measurements that any hospital already takes. The module needs no new test and no new equipment.

05 Training Process

DIRECT ANSWER

There is no training. Module 2 has no model that learns from examples, so there is no training step, no training data, and no training time. Running the module produces the same answer for the same inputs every time.

5.1 · WHAT WAS DONE INSTEAD — SETTING IT UP

The work that replaces training was choosing and checking the fixed values. These steps were carried out once, in this order.

1. **Implement the published equations.** CKD-EPI 2021 and Du Bois were written out from their published definitions.
2. **Check the implementation against a known answer.** The published reference case for CKD-EPI 2021 is a 50-year-old man with a creatinine of 1.0, whose eGFR should be 92. Our code returns 92. This is an automated test, so it runs every time.
3. **Choose the reference point for normal kidney function.** Set at an eGFR of 90, because that is the lower edge of the FDA's "normal" category — the group a kidney study compares everyone else against.
4. **Adopt the FDA category boundaries.** 90, 60, 30 and 15. Taken from the guidance rather than chosen by us.
5. **Choose the exposure bands.** 1.25× and 2×, following standard regulatory practice for what counts as no meaningful change and what counts as clinically important.
6. **Choose the study design cut-off.** A drug is treated as mainly kidney-removed when `fe` is 0.30 or above. This one is our decision and is labelled as such in the code.
7. **Write the liver function scoring.** Built from three routine blood tests, plus an age adjustment. This is labelled in the code as giving a sensible ordering of patients rather than validated absolute values.
8. **Check the whole thing against real data.** Described in Part 6. Nothing was adjusted afterwards to make those numbers look better.

5.2 · HOW TO REPRODUCE IT

```
.venv\Scripts\activate
pip install -r requirements.txt

python smoke_test.py          # runs the automated checks, including Module 2
streamlit run app.py          # opens the web interface
```

There is no training command, because there is nothing to train. The numbers in Part 6 are produced by the validation code and can be regenerated at any time.

06 Validity & Accuracy

6.1 · HOW THE CHECKING WAS DONE

Module 2's prediction was compared against 25 real observations covering 17 drugs, taken from FDA labels and published studies. For each one, the module was given only the drug's `fe` value and the patients' kidney function, and asked to predict the exposure change. That prediction was compared with what the source reported.

Predictions are scored the way this kind of model is scored in published literature: by **fold error**, meaning how many times too high or too low the prediction was. A fold error of 1.0 is exact. The usual acceptance bar for this class of model is a factor of 2.

6.2 · HOW CLOSE THE NUMBERS ARE

MEASURE	RESULT	MEANING
Within 2×	88%	Clears the usual acceptance bar
Within 1.5×	64%	
Within 1.25×	40%	
Average fold error	1.46	Typically off by about 46%
Bias	0.74	Below 1 means the module under-predicts : it says the drug builds up less than it really does, by about 26% on average
Correct exposure band	65%	Whether prediction and reality fall in the same band (no change / adjust / meaningful)
Meaningful changes missed	7 of 13	Cases where reality showed a 2×-or-more change and the module predicted less
"No change" statements matched	9 of 9	Where a label stated plainly that kidney disease changes nothing, the module agreed every time

6.3 · HOW OFTEN THE DECISION IS RIGHT

The exposure figure is an intermediate step. The output that gets acted on is the study design recommendation, and it is measured separately.

DRUGS ASSESSED	DESIGN CORRECT	MISSED FULL STUDY	UNNECESSARY FULL STUDY
17	16 of 17	0 the costly error	1 the cheap error

THE RESULT WORTH STATING

Across 17 drugs with real data, the module never once recommended a reduced study when the evidence called for a full one. Its single error was the opposite kind — recommending a full study for one drug that did not need it, which wastes part of one study rather than allowing a problem through.

6.4 · WHY THE NUMBERS IN 6.2 AND 6.3 LOOK INCONSISTENT

The module under-predicts, yet gets the decision right almost every time. That is not a contradiction, and the reason is worth understanding.

The decision depends on which side of a threshold the answer falls, not on the exact figure. A prediction can be 26% too low and still land on the correct side. The prediction is also only one of two triggers for a full study — a drug with `fe` of 0.30 or above gets a full study regardless of the predicted number. That second rule catches cases the prediction alone would miss.

6.5 · WHY IT UNDER-PREDICTS

The reason is written in the module's own documentation. The equation assumes that when the kidneys fail, everything else keeps working normally. In reality, kidney failure also slows the liver's ability to break down drugs. The equation cannot see that, so it predicts less build-up than actually happens.

This is a known missing piece with a named cause, not unexplained error. It is stated rather than hidden because the direction is safe to state: the module errs toward under-estimating build-up, and the decision layer is built to tolerate that.

6.6 · AUTOMATED TESTS

Module 2 is covered by automated checks that run with `python smoke_test.py` :

- CKD-EPI 2021 returns the published reference value of 92.
- Higher creatinine always produces a lower eGFR.
- Body size adjustment works in the right direction — a smaller body gives a lower absolute figure.
- A healthy average adult shows unchanged exposure, not inflated exposure.
- Exposure figures never run away to impossible values.
- The study design rule picks sensibly across known drugs.
- A healthy control patient generates no kidney study recommendations.
- Every scored observation has a traceable source link.
- The count of missed full studies is asserted to be zero, so a future change cannot quietly reintroduce that error.

6.7 · THE INDEPENDENT SIMULATION CHECK

A second check compares the equation against whole-body simulation in PK-Sim. This work is **in progress and not yet complete**.

The first attempt produced numbers that had to be discarded. The simulations had been built without their metabolism and transport components attached, so they described a drug removed entirely by the kidney rather than the real drug. They are being rebuilt correctly. The results are excluded from this report rather than reported with a caveat.

A LIMIT FOUND DURING THAT WORK

PK-Sim's kidney disease model accepts eGFR values from 0 to 60 only, because chronic kidney disease is defined as eGFR below 60. The "Mild" category, which sits between 60 and 89, therefore cannot be checked by simulation at all. This is a property of the published model, not of our work, and it is stated openly.

07 Dependencies

DEPENDS ON	FOR WHAT	WHY IT MATTERS
Module 1	The starting risk level and score for a drug pair	Module 2 adjusts Module 1's answer for a specific patient. It does not decide whether two drugs interact — that is Module 1's job. If Module 1's starting value is wrong, Module 2 adjusts a wrong number
trialsense/drugs.py	The fe value and the yes/no flags for each drug	This is the single most influential input. The study design recommendation for a drug is driven mainly by its fe
data/renal_validation.json	The real observations used for checking	Needed only for validation, not to run the module
Python, Streamlit, Plotly	Running the code and showing the result	Standard, free, and installed from requirements.txt
PK-Sim, R, ospsuite, rSharp	The independent simulation check	Optional. Not required to run the product

7.1 • WHAT DOES NOT DEPEND ON MODULE 2

Module 3, which handles antibiotic resistance, is entirely separate. It shares no data and no code with Module 2. A failure in one does not affect the other.

08 Limitations

8.1 · ACCURACY LIMITS

- **It under-predicts.** On average it says drugs build up about 26% less than they really do.
- **It missed 7 of 13 meaningful changes.** Where real data showed a 2×-or-greater rise, the module predicted a smaller rise in more than half of those cases.
- **The exposure band is right about 65% of the time.** The decision built on top of it is right 16 times out of 17, but the underlying number is less reliable than that figure suggests.

8.2 · LIMITS OF THE INPUTS

- `fe` is a single rounded number per drug. In reality it varies with dose, formulation and the patient's genetics.
- The liver function score is built from three blood tests. The code states plainly that it gives a sensible ordering of patients but that the absolute values are not validated.
- eGFR is an estimate calculated from a blood test, not a direct measurement of kidney function.
- The source studies report kidney function in different units and often as a range. Each group was collapsed to one representative figure, which adds error on top of the module's own.

8.3 · WHAT THE MODEL DOES NOT REPRESENT AT ALL

- Transporter proteins that actively pump drugs out of the body.
- Changes in how tightly a drug binds to blood proteins.
- Dialysis, which removes drug by a route the equation has no term for.
- Genetic differences between individuals in drug-processing enzymes.
- Interactions involving more than two drugs at once.

8.4 · POPULATION LIMITS

BE PRECISE ABOUT THIS ONE

Module 2 has **not** been validated on Indian patient data, and no such dataset is used. The equations are population-agnostic: CKD-EPI 2021 does not take race as an input, and the validation sources are mostly non-Indian. The module does respond to body size, so a smaller patient is predicted to accumulate more of a kidney-removed drug — but that is a consequence of the equation, not a finding validated against Indian measurements.

Module 3 is the part of the project that uses genuine Indian data. The two claims should not be merged.

8.5 · SCOPE LIMITS

- 39 drugs, chosen for demonstration coverage. It is not a complete formulary.
- The validation covers 17 of those 39. The other 22 have no usable published number.
- The study design rule follows the structure of the FDA guidance but is our own interpretation, not an approved method.
- This is a research screening prototype. It is not a diagnostic tool and not a clinical decision-support system.

09 Future Scope

9.1 • HIGHEST VALUE — ADD THE MISSING MECHANISM

The equation currently assumes the liver keeps working normally when the kidneys fail. Adding a term that reduces non-kidney removal as kidney function falls should correct the under-prediction, because it targets the known cause.

The important condition: the shape of that term must be worked out from a source other than the 25 validation observations. If it were fitted to those observations and then scored against them, the resulting accuracy figure would be meaningless. The simulation work in section 6.7 is intended to supply that independent source.

9.2 • IMPROVE THE MOST INFLUENTIAL INPUT

Because the answer depends heavily on fe , and fe is currently a rounded textbook value, replacing it with a sourced figure from each drug's regulatory label would improve accuracy without changing any equation.

Testing showed that when fe is varied by 0.10 either way, the study design recommendation changes for **4 of 39 drugs** and stays the same for the other 35. Those four sit near the 0.30 cut-off, so sourcing accurate fe values for them specifically would address most of the fragility.

9.3 • REPORT UNCERTAINTY INSTEAD OF HIDING IT

The study design rule currently returns one of two answers. A third option — marking a drug as borderline when its fe sits close to the cut-off and the value is not from a sourced reference — would turn hidden fragility into a visible flag. This requires no new data.

9.4 • COMPLETE THE SIMULATION CHECK

Rebuild the simulations correctly for three to four drugs spanning a wide range of fe , including at least one barely removed by the kidney as a negative control. Any correction term should show no effect there.

9.5 • EXTEND THE VALIDATION SET

22 of the 39 drugs have no usable published figure. Extending the collection would give a larger test set, and would allow splitting it into one part for development and a separate part for testing — which is what would make a future correction term credible.

9.6 · INDIAN POPULATION WORK

Two things would be needed, in order. First, real Indian body size distributions from published national survey data, replacing a single assumed figure. Second, published Indian studies of drug exposure in kidney impairment, which would be the only way to genuinely validate an Indian claim. The second is a literature collection project and should not be described as done until it is.

10 Conclusion

WHAT HAS BEEN BUILT

Module 2 takes seven routine measurements about a patient and a drug's kidney removal fraction, and returns two things: how much that drug is expected to build up in that patient, and what kind of kidney impairment study the drug needs before trials. It uses four published equations, one equation of our own, and a set of fixed rules. It contains no machine learning and nothing that was trained.

WHAT THE MEASUREMENTS ESTABLISH

WITHIN 2× 88% 25 real observations	AVERAGE FOLD ERROR 1.46	BIAS 0.74 under-predicts	DESIGN CORRECT 16 / 17 drugs	MISSED FULL STUDIES 0
"NO CHANGE" AGREED 9 / 9				

THE PRINCIPAL LIMITATION

STATED ONCE, CLEARLY

The exposure prediction is biased low. It says drugs build up about 26% less than they actually do, and it misses more than half of the large changes. The cause is known and named: the equation holds non-kidney removal at full strength when the kidneys fail, and real kidney failure slows that too.

What survives that limitation is the decision. Across 17 drugs with real data, the recommendation was correct 16 times, and the one error was in the safe direction.

HOW TO DESCRIBE IT HONESTLY

DO NOT SAY	SAY INSTEAD
"Module 2 uses AI"	"Module 2 is published equations and rules. Module 3 is where the machine learning is"
"Validated on Indian patients"	"Population-agnostic equations, validated on FDA label data. No Indian PK dataset"
"88% accurate"	"88% of predictions fall within a factor of 2, the usual bar for this kind of model"
"It can waive kidney studies"	"It recommends which design to use. It is a starting point for a regulator conversation, not an approval"
"We saved X crore"	"A reduced study recruits about half as many patients. We have not run a real programme through this"

A MEASUREMENT WORTH RECORDING

The most useful finding from building this module is not the accuracy figure. It is that a prediction can be systematically wrong in a known direction and still support a correct decision, provided the two are measured separately. Reporting a single combined accuracy score would have concealed both halves of that result.

TrialSense is a research screening prototype for pharmaceutical research and development. It is not a diagnostic tool and not a clinical decision-support system. No output should be used to treat anyone.

Figures in Part 6 are produced by `trialsense/pk_validation.py` and can be regenerated at any time. Automated checks run with `python smoke_test.py`.